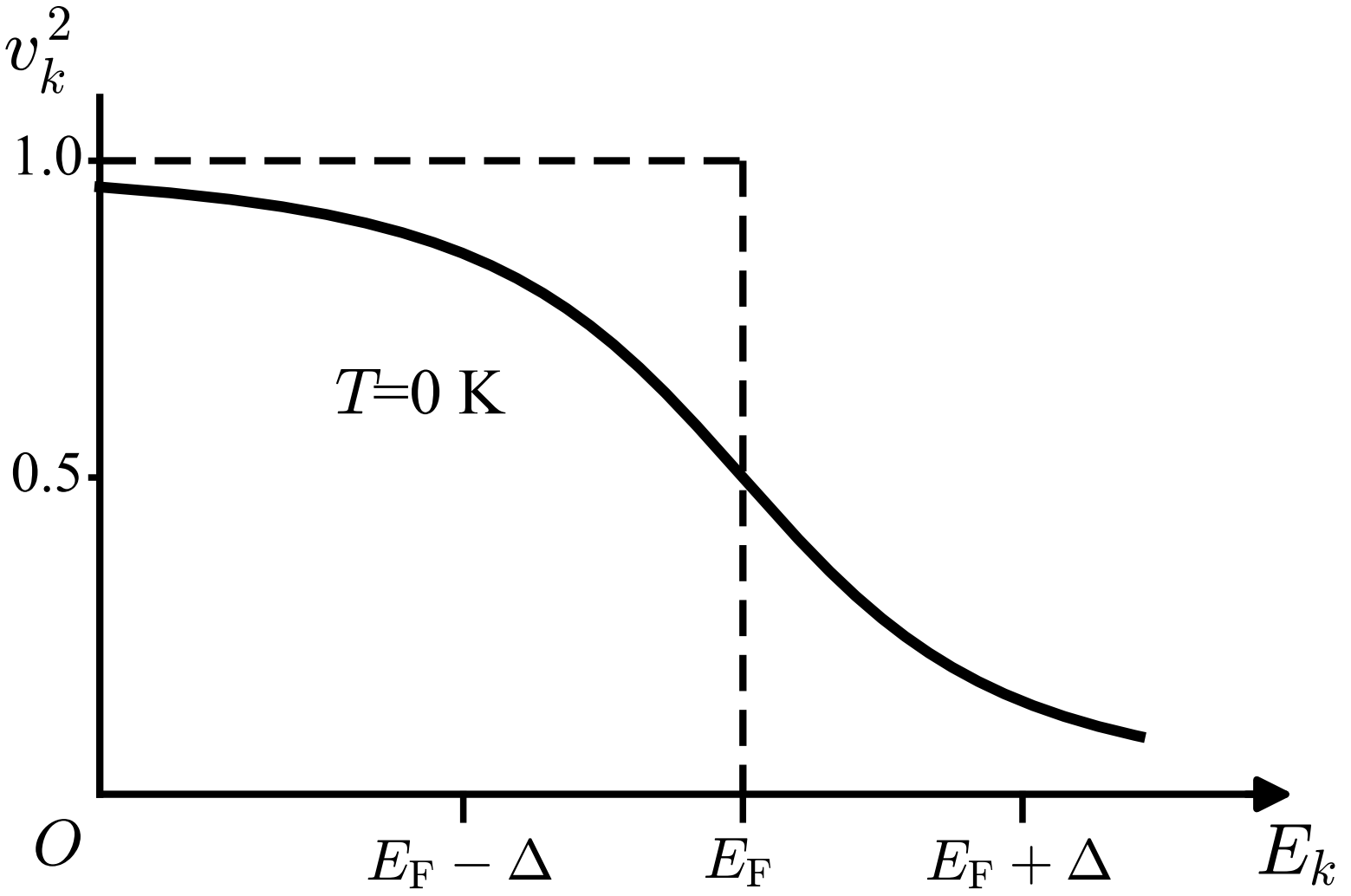

$$v_k^2 = \frac{1}{2}\left\{1 - \frac{(E_k - E_{\mathrm{F}})}{\sqrt{(E_k - E_{\mathrm{F}})^2 + \Delta^2}}\right\}$$